

Comparative Analysis of Hotjar Session Recordings

1 Data structure and quality

Two CSV files covering the last three months of session recordings were provided:

Dataset	Traffic source	Records	Notes
Dataset 1	Forbes SEM (search-engine marketing) campaigns	5 162 sessions	traffic arrives via paid ads on forbes.com
Dataset 2	Finance Advisors referrals	13 559 sessions	traffic arrives via the retirement.financeadvisors.com referral programme

Both datasets share the same schema. Each row corresponds to a single Hotjar session and includes 24 columns such as recording URL, timestamp, user status (new/returning), device, browser and operating system, entry and exit page, activity metrics (action count, page count and duration), frustration signals (rage clicks and U-turns) and a comma-separated **Events** column listing triggered UI events. The presence of identical columns means the datasets can be directly compared. Missing values are minimal: most rows have values for the core metrics, although some sessions lack events or U-Turn counts.

Key Hotjar definitions used in this analysis:

- **Average session duration** measures the length of time a user stays on the site from landing until they leave or become inactive for 30 minutes ¹.
- **Average pages per session** represents the mean number of pages viewed in a session and indicates how “sticky” a site is ².
- **Rage clicks** occur when users repeatedly click on a specific part of a page in a short period. This behaviour signals frustration and is often caused by slow pages, confusing UI or broken elements ³.
- **U-turns** are when a user clicks into a new area of the site and immediately returns to the previous page, suggesting difficulty finding what they need ⁴.

2 User-behaviour comparison

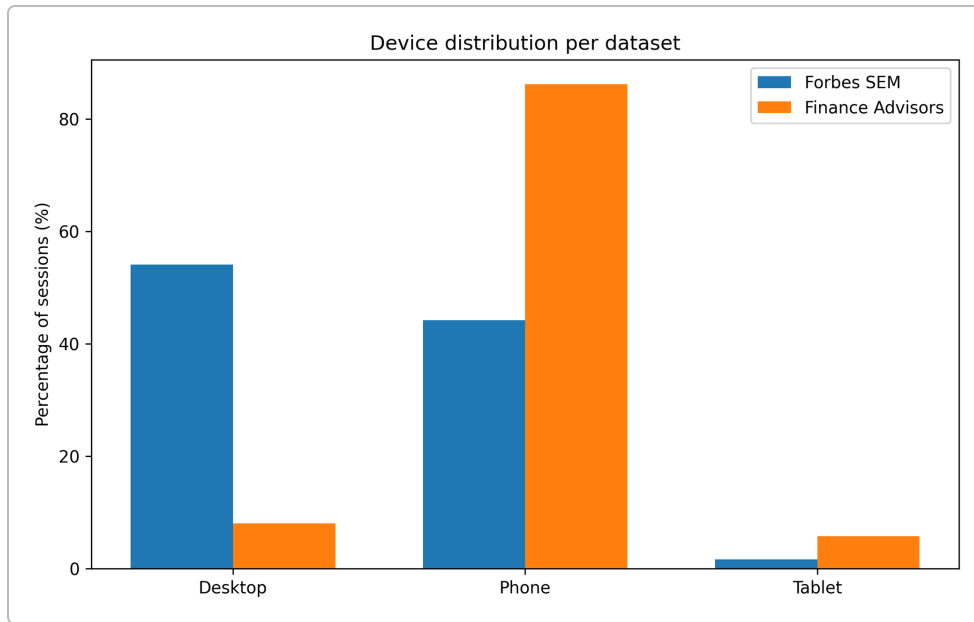
2.1 Engagement and navigation metrics

Metric	Forbes SEM (Dataset 1) – mean	Finance Advisors (Dataset 2) – mean	Significance (two-sample t-test)	Interpretation
Pages per session	10.51 pages	13.22 pages	$p \approx 3.9 \times 10^{-68}$	Finance Advisor users navigate through ~26 % more pages, indicating a longer or more complex flow.
Session duration	182.3 s (≈ 3.0 min)	220.4 s (≈ 3.7 min)	$p \approx 3.7 \times 10^{-20}$	Finance users spend significantly longer on the site.
Actions (clicks, scrolls, taps)	50.86	57.30	$p \approx 3.6 \times 10^{-15}$	Finance sessions contain more user interactions.
Rage clicks	0.0008 per session	0.0018 per session	$p \approx 0.34$	Very rare in both segments; difference not significant.
U-turns	0.109	0.194	$p \approx 3.8 \times 10^{-20}$	Finance sessions contain ~78 % more U-turns, implying more confusion.
Time per page (duration ÷ pages)	24.34 s/page	20.42 s/page	$p \approx 1.1 \times 10^{-4}$	Forbes visitors spend more time on each page, possibly reading long content.
Actions per second	0.35	0.41	$p \approx 4.5 \times 10^{-11}$	Finance visitors interact more frequently during their visit.

The differences in pages per session, duration, actions, U-turns, time per page and actions per second are statistically significant ($p < 0.05$), meaning they likely reflect genuine behavioural differences rather than noise. Only the extremely low rage-click counts show no significant difference between groups.

2.2 Device and technology usage

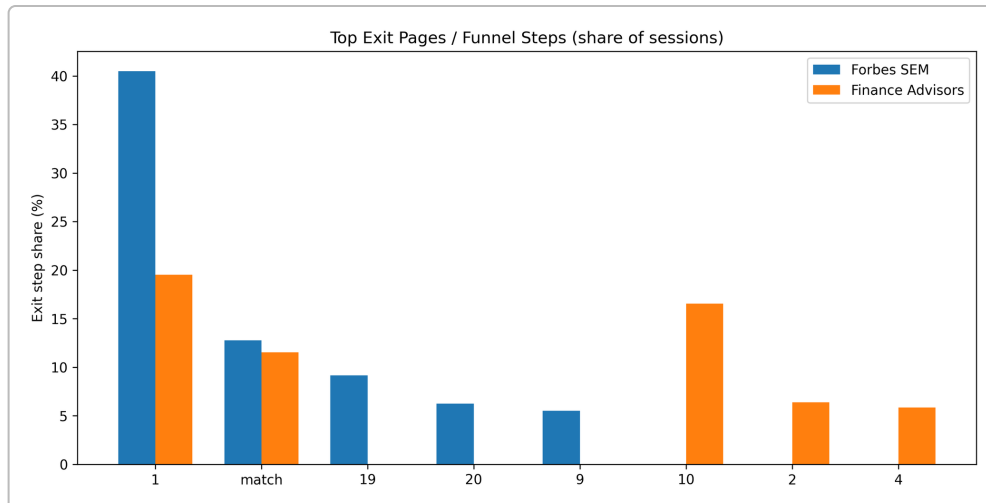
Finance Advisors visitors predominantly use mobile devices. ~86 % of Finance sessions come from **phones**, compared with 44 % for Forbes. Desktop traffic represents over half of Forbes sessions but less than 8 % of Finance sessions. Tablets are a very small share for both groups. The bar chart below summarises the device distribution.



Both audiences overwhelmingly access the site from the United States (US). The Finance dataset also contains small clusters of users from the United Kingdom and Netherlands, whereas Forbes has a handful of sessions from India, the Dominican Republic and Greece. More than 98 % of sessions in each dataset are from **new** users; repeat visitors are rare.

2.3 Navigation patterns and conversion funnel

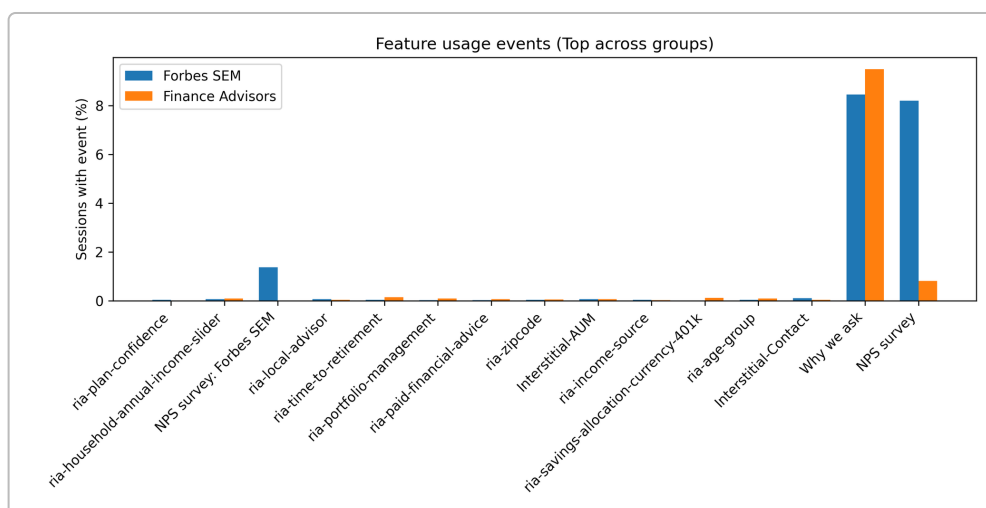
Because Hotjar exports only include the landing and exit pages rather than full click paths, navigation analysis focuses on the exit step number embedded in the URL. Pages with numeric slugs correspond to steps in the onboarding questionnaire; `/match` pages represent successful advisor matches or conversions. The figure below shows the top exit pages and their share of sessions.



- **Early drop-off:** 40 % of Forbes visitors exit on step 1 of the questionnaire, versus ~20 % for Finance visitors. In the Finance flow, the largest drop-off (16.5 % of sessions) occurs at **step 10**, suggesting a bottleneck around this stage.
- **Late-stage drop-off:** Forbes sessions show exits at steps 19–21. These steps only appear in the Forbes SEM flow; they may represent extended forms or eligibility questions that discourage completion.
- **Conversion rate:** Sessions ending on a `/match` page are treated as conversions. Forbes SEM users have a slightly higher conversion rate ($\approx 12.8\%$) than Finance Advisors users ($\approx 11.5\%$).

2.4 Feature usage

The **Events** column lists triggered UI events, which include expansions of help sections (e.g., `why_we_ask_expanded`), Net Promoter Score (NPS) survey triggers and answers to questions in the “Retirement Income Advisor” (RIA) flow. The chart below compares the frequency of the most common non-publisher events. Percentages are calculated relative to the total number of sessions in each dataset.



Key observations:

- **Why we ask explanation:** The most common event in both datasets is expansion of the “Why we ask” help text. ~8.3 % of Forbes sessions and 9.1 % of Finance sessions trigger this, indicating that users need reassurance about why personal information is requested.
- **NPS survey:** The NPS survey appears in ~8.1 % of Forbes sessions and 0.8 % of Finance sessions. The difference suggests that the survey may be targeted to SEM traffic.
- **Question events:** Finance visitors trigger far more RIA question events (e.g., time-to-retirement, savings allocation, household income slider). These events appear in up to 0.3 % of Finance sessions but are almost absent in Forbes sessions, implying that Finance visitors proceed deeper into the questionnaire.
- **Interstitial forms:** Forbes sessions occasionally include `Interstitial-Contact` and `Interstitial-AUM` events. These pop-ups may present friction, as the corresponding questions have low completion rates.

3 Audience insights

3.1 User intent differences

Forbes SEM audience: Traffic from paid ads likely includes users who are casually exploring investment options. They tend to use desktops, view fewer pages and exit early. Event logs show limited engagement with retirement-planning questions. The high time-per-page figure suggests they may be reading explanatory text or terms carefully before deciding whether to continue. Early drop-off at step 1 and low completion of interstitial forms indicate that these visitors may be sceptical about sharing personal information or unsure what the process entails.

Finance Advisors audience: Referral traffic appears to consist of more motivated retirees or pre-retirees. These users predominantly browse on mobile, move quickly through pages, trigger more question events and spend longer overall. The large number of U-turns and the drop-off at step 10 imply friction points in the questionnaire—perhaps a particularly complex or unclear question. Events such as `ria-time-to-retirement`, `ria-savings-allocation-currency-401k`, `ria-portfolio-management` and `ria-age-group` suggest users are providing detailed financial information to receive personalised advice.

3.2 Behavioural characteristics

- **Engagement intensity:** Finance visitors perform more actions per second and more pages per session, indicating a faster pace and higher engagement. Forbes visitors are slower and less interactive.
- **Device differences:** Forbes SEM traffic is mixed between desktop and mobile, while Finance traffic is overwhelmingly mobile. Mobile users may be more prone to U-turns if the interface is not optimised for small screens.
- **Frustration signals:** U-turns occur nearly twice as often in Finance sessions (0.194 per session vs 0.109). Since a U-turn indicates a user clicked into a new area and immediately went back ⁴, this suggests confusion or poorly labelled navigation for Finance visitors.
- **Information needs:** The popularity of the “Why we ask” explanation across both groups highlights a universal need to build trust and clarify why sensitive questions are necessary.

3.3 Pain points

- **Early-stage abandonment:** Step 1 of the Forbes flow loses 40 % of sessions, pointing to a problematic landing question or copy.
- **Mid-flow bottleneck:** Finance users are most likely to drop off at step 10. This step may require detailed financial data; addressing clarity or complexity here could reduce abandonment.
- **U-turn spikes:** Elevated U-turn counts for Finance sessions suggest navigation or labelling issues on certain pages. Users might be clicking the wrong option and returning immediately.
- **Lack of cross-device optimisation:** Because Finance traffic is largely mobile, any desktop-first design may cause friction and U-turns.

4 Key differentiators and statistical significance

The most significant differences between the audiences include:

1. **Device usage** – 86 % of Finance visitors use phones versus 44 % of Forbes visitors. This difference requires distinct design considerations.
2. **Navigation depth** – Finance users visit about 13 pages per session and progress further into the questionnaire, whereas Forbes visitors average ~10 pages and often exit at the first step. The difference is highly significant ($p \approx 3.9 \times 10^{-68}$).
3. **Session duration and pace** – Finance sessions are longer overall (220 s vs 182 s, $p \approx 3.7 \times 10^{-20}$) but shorter per page (20 s vs 24 s). Forbes visitors read longer; Finance visitors skim faster and interact more often.
4. **U-turn frequency** – Finance sessions include nearly twice as many U-turns ($p \approx 3.8 \times 10^{-20}$), signalling more confusion in the referral audience.
5. **Feature usage** – Events like NPS surveys and interstitial pop-ups appear mostly in the Forbes SEM flow. Finance users answer more detailed retirement questions.

Across most metrics the differences are statistically significant ($p < 0.05$), so they likely reflect real behavioural variation rather than sampling noise. The only exceptions are rage clicks and actions per page, where differences are negligible.

5 Strategic recommendations

5.1 Audience-specific optimisations

Forbes SEM users (desktop-heavy, cautious)

Priority	Recommendation	Rationale & impact
High	Simplify the initial question (step 1) and clarify value	40 % of Forbes visitors exit on the first step. Re-write copy to emphasise benefits and reduce the amount of information requested up front. A/B testing could increase the ~12.8 % conversion rate by several percentage points.

Priority	Recommendation	Rationale & impact
High	Reduce interstitial friction	Interstitial contact and AUM questions are rare and may cause abandonment. Consider integrating these questions later in the flow or making them optional to reduce cognitive load.
Medium	Targeted NPS survey	The NPS survey currently triggers in ~8 % of sessions. Use behaviour-based triggers (e.g., after completing the flow) to gather more relevant feedback without interrupting new users.
Low	Optimise desktop UI	Over half of SEM traffic uses desktop; ensure large-screen layouts feel polished. Provide contextual help to build trust in the process and reduce the need for the “Why we ask” expansion.

Finance Advisors users (mobile-dominant, engaged)

Priority	Recommendation	Rationale & impact
High	Streamline step 10 and subsequent questions	Step 10 is the biggest drop-off point. Simplify language, break complex questions into smaller chunks or provide examples. Reducing friction here could lift conversion by ~1–2 percentage points.
High	Improve mobile usability and navigation	With 86 % of visits on phones and elevated U-turn counts, ensure buttons and menus are clearly labelled and sized appropriately. Adding progress indicators and back buttons can reduce confusion.
Medium	Contextual explanations for sensitive questions	Continue using the “Why we ask” feature and expand explanations near complicated questions (e.g., financial data requests) to maintain trust.
Low	Personalise question order based on referral source	If the user originates from a retirement advice article, pre-populate or reorder questions to reflect their intent (e.g., emphasise time-to-retirement early).

5.2 Universal improvements

Priority	Recommendation	Rationale
High	Provide a clear value proposition and progress indicator	Both groups show early drop-off, indicating uncertainty about the benefit or length of the questionnaire. Display a progress bar and a concise statement about what users will receive upon completion (e.g., “Find your ideal advisor in 3 minutes”).

Priority	Recommendation	Rationale
Medium	Strengthen trust messaging	The popularity of the “Why we ask” explanation reveals that users are hesitant to share personal information. Include privacy assurances and links to data policies up front.
Medium	Conduct usability testing on critical steps	Use session replays and Hotjar’s rage-click and U-turn filters   to watch sessions where users struggle. Identify whether issues stem from wording, layout or technical problems.
Low	Optimize for lower-bandwidth users	A small number of sessions from India and other countries may experience slower page loads. Compress assets and ensure forms remain responsive on low-end devices.

5.3 Prioritised action plan and business impact

Priority	Initiative	Expected impact	Resources & timeline
High	Simplify step 1 (Forbes)	Increase completion of early steps and raise conversion rate from 12.8 % toward Finance levels. A small copy-editing project, A/B tested over 2–3 weeks.	Product copywriter & designer; minimal development.
High	Redesign step 10 (Finance)	Reduce drop-off at step 10; potential improvement of 1–2 percentage points in conversion. Requires UX design and possibly backend changes; ~4 weeks.	UX designer, frontend developer and QA.
High	Mobile navigation overhaul	Lower U-turn counts and improve satisfaction for mobile-heavy Finance users. Could increase retention and cross-selling opportunities. 6–8 weeks.	Design and engineering resources for responsive layout updates.
Medium	Trust and privacy messaging	Build user confidence across both audiences and reduce early exits. Implement privacy badges, short copy near forms and FAQs. 2 weeks.	Marketing/UX writing.
Medium	Progress indicators and value proposition	Set clear expectations; reduce abandonment; apply to both audiences. 2–3 weeks.	Front-end development.
Low	Expand contextual help	Extend the “Why we ask” feature to more fields; create tooltips or links to resources explaining financial terminology. Ongoing.	Product and content teams.

6 Business impact and ROI

Implementing these recommendations is expected to improve conversion rates, user satisfaction and, ultimately, revenue:

- **Conversion uplift:** Simplifying step 1 for Forbes and streamlining step 10 for Finance could each improve conversion by 1–2 percentage points. Given current conversion rates (~12.8 % and 11.5 %), even a 1 percentage-point increase translates to hundreds of additional advisor matches per month.
- **Customer acquisition costs:** Optimising the SEM flow reduces wasted ad spend by retaining more of the expensive paid clicks. Improving the referral flow increases lifetime value by connecting more qualified retirees to advisors.
- **Development effort:** High-impact changes involve copy updates, interface redesign and responsive improvements. These can be rolled out incrementally over 2–8 weeks, balancing quick wins with longer-term refinements.
- **ROI and timeline:** Because the recommended changes directly target abandonment points, the payback period is short. Early copy optimisations may yield results within a month, while structural mobile improvements will deliver sustained benefits over the next quarter.

By focusing on the distinct needs of SEM and referral audiences and addressing universal pain points, the customer-experience team can enhance trust, reduce friction and drive more users to complete the advisor-matching process.

1 2 Hotjar Dashboards: How to Turn Its Insights into Action

<https://www.hotjar.com/dashboards/how-to-use/>

3 How to Use Rage Clicks To Improve User Experience

<https://www.hotjar.com/blog/rage-clicks/>

4 13 Behavior Analytics Metrics To Track + Analyze

<https://contentsquare.com/guides/behavior-analytics/metrics/>